Chapter 6. Differential Forms

Multilinear algebra, alternating tensors and the wedge product, differential forms, the operator d, and pullbacks

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Overview

Vector integral calculus (Green, Gauss, Stokes) on k-manifolds in $\mathbb{R}^n$ needs *multilinear algebra* (§26–§28, from Chapter 1 only). Combined with differentiation it gives *differential forms* and their operators, replacing scalar and vector fields and grad, curl, div; integration and Stokes' theorem follow in Chapter 7.

§26. Multilinear Algebra

Motivation. An integrand over a k-surface is a function of k tangent vectors, linear in each: a tensor.

26.1 Tensors

Definition 26.1 (k-tensor)

Let V be a vector space and $V^k = V \times \cdots \times V$ the set of k-tuples $(\mathbf{v}_1, \ldots, \mathbf{v}_k)$ of vectors of V. A function $f : V^k \to \mathbb{R}$ is **linear in the ith variable** if, for fixed $\mathbf{v}_j$ $(j \neq i)$, the function $T(\mathbf{v}) = f(\mathbf{v}_1, \ldots, \mathbf{v}_{i-1}, \mathbf{v}, \mathbf{v}_{i+1}, \ldots, \mathbf{v}_k)$ is linear. f is **multilinear** if it is linear in the ith variable for each i; it is then called a **k-tensor**, or a **tensor of order k**, on V. The set of k-tensors on V is $\mathcal{L}^k(V)$. $\mathcal{L}^1(V)$ is the set of linear maps $V \to \mathbb{R}$, the **dual space** V^* of V.

Theorem 26.1

The set of all k-tensors on V constitutes a vector space if we define

$$(f+g)(\mathbf{v}_1, \ldots, \mathbf{v}_k) = f(\mathbf{v}_1, \ldots, \mathbf{v}_k) + g(\mathbf{v}_1, \ldots, \mathbf{v}_k), \qquad (cf)(\mathbf{v}_1, \ldots, \mathbf{v}_k) = c(f(\mathbf{v}_1, \ldots, \mathbf{v}_k)).$$

Proof. Sums and scalar multiples of functions linear in each variable are linear in each variable; the vector space axioms hold because they hold for real numbers pointwise. The zero tensor is the function whose value on every k-tuple is zero. ■

Lemma 26.2

Let $\mathbf{a}_1, \ldots, \mathbf{a}_n$ be a basis for V. If $f, g : V^k \to \mathbb{R}$ are k-tensors on V, and if $f(\mathbf{a}_{i_1}, \ldots, \mathbf{a}_{i_k}) = g(\mathbf{a}_{i_1}, \ldots, \mathbf{a}_{i_k})$ for every k-tuple $I = (i_1, \ldots, i_k)$ of integers from the set $\{1, \ldots, n\}$, then $f = g$.

(The integers $i_1, \ldots, i_k$ need not be distinct or in any order.)

Proof. Write $\mathbf{v}_i = \sum_{j=1}^n c_{ij}\mathbf{a}_j$. Expanding by multilinearity, one variable at a time,

$$f(\mathbf{v}_1, \ldots, \mathbf{v}_k) = \sum_{1 \le j_1, \ldots, j_k \le n} c_{1j_1} c_{2j_2} \cdots c_{kj_k} f(\mathbf{a}_{j_1}, \ldots, \mathbf{a}_{j_k}),$$

and likewise for g. ■

Theorem 26.3

Let V be a vector space with basis $\mathbf{a}_1, \ldots, \mathbf{a}_n$. Let $I = (i_1, \ldots, i_k)$ be a k-tuple of integers from the set $\{1, \ldots, n\}$. There is a unique k-tensor ϕ_I on V such that, for every k-tuple

$J = (j_1, \dots, j_k)$ from the set $\{1, \dots, n\}$,

$$\phi_I(\mathbf{a}_{j_1}, \dots, \mathbf{a}_{j_k}) = \begin{cases} 0 & \text{if } I \neq J, \\ 1 & \text{if } I = J. \end{cases} \qquad (*)$$

The tensors ϕ_I form a basis for $\mathcal{L}^k(V)$.

The ϕ_I are the **elementary k-tensors** on V corresponding to the basis $\mathbf{a}_1, \dots, \mathbf{a}_n$. There are n^k of them, so $\dim \mathcal{L}^k(V) = n^k$. For $k = 1$, the basis $\phi_1, \dots, \phi_n$ of V^* is the basis **dual** to $\mathbf{a}_1, \dots, \mathbf{a}_n$.

Proof. Uniqueness follows from Lemma 26.2.

Existence: for $k = 1$, define the linear map ϕ_i by $\phi_i(\mathbf{a}_j) = 0$ $(i \neq j)$, 1 $(i = j)$ (Theorem 1.4). For $k > 1$ set

$$\phi_I(\mathbf{v}_1, \dots, \mathbf{v}_k) = [\phi_{i_1}(\mathbf{v}_1)] \cdot [\phi_{i_2}(\mathbf{v}_2)] \cdots [\phi_{i_k}(\mathbf{v}_k)];$$

it is multilinear and has the values $(*)$. Given a k-tensor f, let $d_I = f(\mathbf{a}_{i_1}, \dots, \mathbf{a}_{i_k})$ and $g = \sum_J d_J \phi_J$ (sum over all k-tuples J). Then $g(\mathbf{a}_{i_1}, \dots, \mathbf{a}_{i_k}) = d_I$, so $f = g$ by Lemma 26.2; the representation is unique by the same lemma. ■

Thus a k-tensor may be defined by specifying its values arbitrarily on k-tuples of basis elements.

Example 26.1

Let $V = \mathbb{R}^n$ with the usual basis and dual basis $\phi_1, \dots, \phi_n$. Then $\phi_i(\mathbf{x}) = \phi_i(x_1\mathbf{e}_1 + \cdots + x_n\mathbf{e}_n) = x_i$: ϕ_i is projection onto the ith coordinate. More generally $\phi_I(\mathbf{x}_1, \dots, \mathbf{x}_k) = \phi_{i_1}(\mathbf{x}_1) \cdots \phi_{i_k}(\mathbf{x}_k)$; writing $X = [\mathbf{x}_1 \cdots \mathbf{x}_k]$ with entries x_{ij},

$$\phi_I(\mathbf{x}_1, \dots, \mathbf{x}_k) = x_{i_1 1} x_{i_2 2} \cdots x_{i_k k},$$

a *monomial* in the components of the $\mathbf{x}_j$. The general k-tensor on $\mathbb{R}^n$ is a linear combination of such monomials: the general 1-tensor is $f(\mathbf{x}) = d_1 x_1 + \cdots + d_n x_n$, the general 2-tensor is $g(\mathbf{x}, \mathbf{y}) = \sum_{i,j=1}^n d_{ij} x_i y_j$, and so on.

26.2 The tensor product

Definition 26.2 (tensor product)

Let f be a k-tensor and g an l-tensor on V. The $(k+l)$-tensor $f \otimes g$ is

$$(f \otimes g)(\mathbf{v}_1, \dots, \mathbf{v}_{k+l}) = f(\mathbf{v}_1, \dots, \mathbf{v}_k) \cdot g(\mathbf{v}_{k+1}, \dots, \mathbf{v}_{k+l}).$$

It is multilinear, and is the **tensor product** of f and g.

Theorem 26.4

Let f, g, h be tensors on V. Then the following properties hold:

(1) (Associativity) $f \otimes (g \otimes h) = (f \otimes g) \otimes h$.

(2) (Homogeneity) $(cf) \otimes g = c(f \otimes g) = f \otimes (cg)$.

(3) (Distributivity) Suppose f and g have the same order. Then $(f+g) \otimes h = f \otimes h + g \otimes h$ and $h \otimes (f+g) = h \otimes f + h \otimes g$.

(4) Given a basis $\mathbf{a}_1, \dots, \mathbf{a}_n$ for V, the corresponding elementary tensors ϕ_I satisfy $\phi_I = \phi_{i_1} \otimes \phi_{i_2} \otimes \cdots \otimes \phi_{i_k}$, where $I = (i_1, \dots, i_k)$.

No parentheses are needed in (4) by associativity; nothing is said about commutativity, which almost never holds.

Proof. Straightforward. For instance, if f, g, h have orders k, l, m, both $(f\otimes(g\otimes h))(\mathbf{v}_1,\dots,\mathbf{v}_{k+l+m})$ and $((f\otimes g)\otimes h)(\mathbf{v}_1,\dots,\mathbf{v}_{k+l+m})$ equal $f(\mathbf{v}_1,\dots,\mathbf{v}_k)\cdot g(\mathbf{v}_{k+1},\dots,\mathbf{v}_{k+l})\cdot h(\mathbf{v}_{k+l+1},\dots,\mathbf{v}_{k+l+m})$. Property (4) is the formula in the proof of Theorem 26.3. ■

26.3 The action of a linear transformation

Definition 26.3 (dual transformation)

Let $T: V\to W$ be linear. The **dual transformation** $T^*:\mathcal{L}^k(W)\to\mathcal{L}^k(V)$ (in the opposite direction) is

$$(T^*f)(\mathbf{v}_1,\dots,\mathbf{v}_k)=f(T(\mathbf{v}_1),\dots,T(\mathbf{v}_k)),$$

i.e. T^*f is the composite of $T\times\cdots\times T: V^k\to W^k$ with $f: W^k\to\mathbb{R}$.

T^*f is multilinear since T is linear and f multilinear.

Theorem 26.5

Let $T: V\to W$ be a linear transformation; let $T^*:\mathcal{L}^k(W)\to\mathcal{L}^k(V)$ be the dual transformation. Then:
(1) T^* is linear.
(2) $T^*(f\otimes g)=T^*f\otimes T^*g$.
(3) If $S: W\to X$ is a linear transformation, then $(S\circ T)^*f=T^*(S^*f)$.

Proof. Straightforward; e.g. for (1),

$$(T^*(af+bg))(\mathbf{v}_1,\dots,\mathbf{v}_k)=(af+bg)(T(\mathbf{v}_1),\dots,T(\mathbf{v}_k))=aT^*f(\mathbf{v}_1,\dots,\mathbf{v}_k)+bT^*g(\mathbf{v}_1,\dots,\mathbf{v}_k).$$

For (2), both sides evaluated at $(\mathbf{v}_1,\dots,\mathbf{v}_{k+l})$ give $f(T\mathbf{v}_1,\dots,T\mathbf{v}_k)\,g(T\mathbf{v}_{k+1},\dots,T\mathbf{v}_{k+l})$. For (3), both sides give $f(S(T(\mathbf{v}_1)),\dots,S(T(\mathbf{v}_k)))$. ■

Property (3) says the diagram $V\xrightarrow{T}W\xrightarrow{S}X$ induces $\mathcal{L}^k(X)\xrightarrow{S^*}\mathcal{L}^k(W)\xrightarrow{T^*}\mathcal{L}^k(V)$ with $(S\circ T)^*=T^*\circ S^*$.

§27. Alternating Tensors

Motivation. Alternating tensors change sign when two vectors are swapped and vanish on degenerate parallelepipeds; the determinant is the model.

27.1 Permutations

Definition 27.1 (permutation, symmetric group)

Let $k\geq 2$. A **permutation** of $\{1,\dots,k\}$ is a one-to-one map σ of this set onto itself; the set of them is S_k. With composition and inverses, S_k is a group, the **symmetric group** (**permutation group**); it has $k!$ elements.

Definition 27.2 (elementary permutation)

For $1 \le i < k$, let $e_i \in S_k$ be defined by $e_i(j) = j$ for $j \neq i, i+1$, $e_i(i) = i+1$, $e_i(i+1) = i$. e_i is an **elementary permutation**; $e_i \circ e_i$ is the identity.

Lemma 27.1

If $\sigma \in S_k$, then σ equals a composite of elementary permutations.

Proof. Say σ *fixes the first* i *integers* if $\sigma(j) = j$ for $1 \le j \le i$ $(0 \le i \le k)$. If $i = k$, σ is the identity, which equals $e_j \circ e_j$. We show: if σ fixes the first $i-1$ integers $(0 < i \le k)$, then $\sigma = \pi \circ \sigma'$ with π a composite of elementary permutations and σ' fixing the first i integers; induction completes the proof. Since σ fixes $1, \dots, i-1$ and is one-to-one, $\sigma(i) = l \ge i$. If $l = i$, take $\sigma' = \sigma$, $\pi =$ identity. If $l > i$, let $\sigma' = e_i \circ \cdots \circ e_{l-1} \circ \sigma$. Then σ' fixes $1, \dots, i-1$ (as σ and $e_i, \dots, e_{l-1}$ do), and $\sigma'(i) = e_i(\cdots(e_{l-1}(l))\cdots) = i$. And $\sigma = e_{l-1} \circ \cdots \circ e_i \circ \sigma'$. ■

Definition 27.3 (sign of a permutation)

Let $\sigma \in S_k$. A pair $i < j$ with $\sigma(i) > \sigma(j)$ is an **inversion** in σ. The **sign** $\operatorname{sgn}\sigma$ is -1 if the number of inversions is odd and $+1$ if it is even; σ is **odd** or **even** accordingly.

Lemma 27.2

Let $\sigma, \tau \in S_k$.

(a) If σ equals a composite of m elementary permutations, then $\operatorname{sgn}\sigma = (-1)^m$.

(b) $\operatorname{sgn}(\sigma \circ \tau) = (\operatorname{sgn}\sigma) \cdot (\operatorname{sgn}\tau)$.

(c) $\operatorname{sgn}\sigma^{-1} = \operatorname{sgn}\sigma$.

(d) If $p \neq q$, and if τ is the permutation that exchanges p and q and leaves all other integers fixed, then $\operatorname{sgn}\tau = -1$.

Proof.

Step 1. For any σ, $\operatorname{sgn}(\sigma \circ e_l) = -\operatorname{sgn}\sigma$. Write the values of σ in order: $(\sigma(1), \dots, \sigma(l), \sigma(l+1), \dots, \sigma(k))$ $(*)$; for $\tau = \sigma \circ e_l$ the sequence is $(\sigma(1), \dots, \sigma(l+1), \sigma(l), \dots, \sigma(k))$ $(**)$. Inversions are pairs of entries appearing in reverse natural order. For $p \neq q$: if neither is l or $l+1$, $\sigma(p), \sigma(q)$ occupy the same slots in both sequences; if one is l or $l+1$ and the other is not, they are in the same relative order in both. Only the pair $\sigma(l), \sigma(l+1)$ changes: it is an inversion in exactly one of the two sequences. So $(**)$ has one more or one fewer inversion than $(*)$.

Step 2. The identity has sign $+1$, and composing with m elementary permutations changes the sign m times: (a). Writing σ, τ as composites of m, n elementary permutations, (b) follows from $(-1)^{m+n} = (-1)^m(-1)^n$.

(c) follows from (b) since $\sigma^{-1} \circ \sigma$ is the identity.

(d) Count inversions: for $p < q$, $q = p + l$, the values of τ in order are $(1, \dots, p-1, q, p+1, \dots, p+l-1, p, p+l+1, \dots, k)$. The pairs $\{q, p+1\}, \dots, \{q, p+l-1\}$ and $\{p+1, p\}, \dots, \{p+l-1, p\}$ are inversions, and so is $\{q, p\}$: $2l-1$ in all, an odd number. ■

27.2 Alternating tensors

Definition 27.4 (symmetric and alternating tensors)

Let f be a k-tensor on V and $\sigma \in S_k$. Define the k-tensor f^σ by

$$f^\sigma(\mathbf{v}_1, \dots, \mathbf{v}_k) = f(\mathbf{v}_{\sigma(1)}, \dots, \mathbf{v}_{\sigma(k)}).$$

f is **symmetric** if $f^e = f$ for each elementary permutation e, and **alternating** if $f^e = -f$ for every elementary permutation e; i.e. f is alternating if

$$f(\mathbf{v}_1, \dots, \mathbf{v}_{i+1}, \mathbf{v}_i, \dots, \mathbf{v}_k) = -f(\mathbf{v}_1, \dots, \mathbf{v}_i, \mathbf{v}_{i+1}, \dots, \mathbf{v}_k) \quad \text{for all } i.$$

The set of alternating k-tensors on V is $\mathcal{A}^k(V)$, a linear subspace of $\mathcal{L}^k(V)$. By convention, $\mathcal{A}^1(V) = \mathcal{L}^1(V)$.

We are primarily interested in alternating tensors.

Example 27.1

Elementary tensors of order $k > 1$ are not alternating, but certain combinations are. With $V = \mathbb{R}^n$ and the dual basis ϕ_i, $f = \phi_{i,j} - \phi_{j,i}$ satisfies

$$f(\mathbf{x}, \mathbf{y}) = x_i y_j - x_j y_i = \det \begin{bmatrix} x_i & y_i \\ x_j & y_j \end{bmatrix},$$

so $f(\mathbf{y}, \mathbf{x}) = -f(\mathbf{x}, \mathbf{y})$. Similarly $g(\mathbf{x}, \mathbf{y}, \mathbf{z}) = \det \begin{bmatrix} x_i & y_i & z_i \\ x_j & y_j & z_j \\ x_k & y_k & z_k \end{bmatrix}$ is an alternating 3-tensor on $\mathbb{R}^n$; $g = \phi_{i,j,k} + \phi_{j,k,i} + \phi_{k,i,j} - \phi_{j,i,k} - \phi_{i,k,j} - \phi_{k,j,i}$. Alternating tensors and determinants are intimately related.

Lemma 27.3

Let f be a k-tensor on V; let $\sigma, \tau \in S_k$.

(a) The transformation $f \to f^\sigma$ is a linear transformation of $\mathcal{L}^k(V)$ to $\mathcal{L}^k(V)$. It has the property that for all σ, τ, $(f^\sigma)^\tau = f^{\tau \circ \sigma}$.

(b) The tensor f is alternating if and only if $f^\sigma = (\operatorname{sgn} \sigma) f$ for all σ. If f is alternating and if $\mathbf{v}_p = \mathbf{v}_q$ with $p \neq q$, then $f(\mathbf{v}_1, \dots, \mathbf{v}_k) = 0$.

Proof. (a) Linearity says $(af + bg)^\sigma = af^\sigma + bg^\sigma$. With $\mathbf{w}_i = \mathbf{v}_{\tau(i)}$,

$$(f^\sigma)^\tau(\mathbf{v}_1, \dots, \mathbf{v}_k) = f^\sigma(\mathbf{w}_1, \dots, \mathbf{w}_k) = f(\mathbf{w}_{\sigma(1)}, \dots, \mathbf{w}_{\sigma(k)}) = f(\mathbf{v}_{\tau(\sigma(1))}, \dots, \mathbf{v}_{\tau(\sigma(k))}) = f^{\tau\circ\sigma}(\mathbf{v}_1, \dots, \mathbf{v}_k).$$

(b) If $f^\sigma = (\operatorname{sgn} \sigma) f$ for all σ, then $f^e = -f$ for elementary e. Conversely, write $\sigma = \sigma_1 \circ \cdots \circ \sigma_m$ with σ_i elementary; by (a), $f^\sigma = ((\cdots (f^{\sigma_m}) \cdots)^{\sigma_2})^{\sigma_1} = (-1)^m f = (\operatorname{sgn} \sigma) f$. If $\mathbf{v}_p = \mathbf{v}_q$ $(p \neq q)$ and τ exchanges p and q, then $f^\tau(\mathbf{v}_1, \dots, \mathbf{v}_k) = f(\mathbf{v}_1, \dots, \mathbf{v}_k)$, while $f^\tau = -f$ since $\operatorname{sgn} \tau = -1$; so $f(\mathbf{v}_1, \dots, \mathbf{v}_k) = 0$. ■

For $k = 1$ nothing is to be done. For $k > n$, $\mathcal{A}^k(V)$ is trivial: some basis element appears twice in each k-tuple of basis elements, so an alternating f vanishes on all of them (Lemma 26.2). For $1 < k \leq n$, an alternating tensor is determined by its values on k-tuples of basis elements with indices in ascending order, which may be specified arbitrarily.

Lemma 27.4

Let $\mathbf{a}_1, \dots, \mathbf{a}_n$ be a basis for V. If f, g are alternating k-tensors on V, and if $f(\mathbf{a}_{i_1}, \dots, \mathbf{a}_{i_k}) = g(\mathbf{a}_{i_1}, \dots, \mathbf{a}_{i_k})$ for every ascending k-tuple of integers $I = (i_1, \dots, i_k)$ from the set $\{1, \dots, n\}$, then $f = g$.

Proof. By Lemma 26.2 it suffices to show f, g agree on an arbitrary $(\mathbf{a}_{j_1}, \dots, \mathbf{a}_{j_k})$. If two indices j_p, j_q coincide, both values are 0 (Lemma 27.3). Otherwise let σ be the permutation such that $I = (j_{\sigma(1)}, \dots, j_{\sigma(k)})$ is ascending. Then $f(\mathbf{a}_{i_1}, \dots, \mathbf{a}_{i_k}) = f^\sigma(\mathbf{a}_{j_1}, \dots, \mathbf{a}_{j_k}) = (\operatorname{sgn}\sigma) f(\mathbf{a}_{j_1}, \dots, \mathbf{a}_{j_k})$, and similarly for g. Since f, g agree on the ascending tuple, they agree on $(\mathbf{a}_{j_1}, \dots, \mathbf{a}_{j_k})$. ∎

Theorem 27.5

Let V be a vector space with basis $\mathbf{a}_1, \dots, \mathbf{a}_n$. Let $I = (i_1, \dots, i_k)$ be an ascending k-tuple from the set $\{1, \dots, n\}$. There is a unique alternating k-tensor ψ_I on V such that for every ascending k-tuple $J = (j_1, \dots, j_k)$ from the set $\{1, \dots, n\}$,

$$\psi_I(\mathbf{a}_{j_1}, \dots, \mathbf{a}_{j_k}) = \begin{cases} 0 & \text{if } I \neq J, \\ 1 & \text{if } I = J. \end{cases}$$

The tensors ψ_I form a basis for $\mathcal{A}^k(V)$. The tensor ψ_I in fact satisfies the formula

$$\psi_I = \sum_\sigma (\operatorname{sgn}\sigma)(\phi_I)^\sigma,$$

where the summation extends over all $\sigma \in S_k$.

The ψ_I are the **elementary alternating k-tensors** on V corresponding to the basis $\mathbf{a}_1, \dots, \mathbf{a}_n$.

Proof. Uniqueness follows from Lemma 27.4. Define ψ_I by the formula. It is alternating: for $\tau \in S_k$,

$$(\psi_I)^\tau = \sum_\sigma (\operatorname{sgn}\sigma)((\phi_I)^\sigma)^\tau = \sum_\sigma (\operatorname{sgn}\sigma)(\phi_I)^{\tau\circ\sigma} = (\operatorname{sgn}\tau)\sum_\sigma (\operatorname{sgn}(\tau\circ\sigma))(\phi_I)^{\tau\circ\sigma} = (\operatorname{sgn}\tau)\psi_I,$$

since $\tau \circ \sigma$ ranges over S_k as σ does. Its values: $\psi_I(\mathbf{a}_{j_1}, \dots, \mathbf{a}_{j_k}) = \sum_\sigma (\operatorname{sgn}\sigma)\phi_I(\mathbf{a}_{j_{\sigma(1)}}, \dots, \mathbf{a}_{j_{\sigma(k)}})$. At most one term is nonzero, the one with $I = (j_{\sigma(1)}, \dots, j_{\sigma(k)})$; as I and J are both ascending this happens only if $I = J$ and σ is the identity, giving value 1.

The ψ_I form a basis: given alternating f, let $d_I = f(\mathbf{a}_{i_1}, \dots, \mathbf{a}_{i_k})$ for ascending I, and $g = \sum_{[J]} d_J \psi_J$ (sum over ascending J). Then f and g agree on ascending k-tuples of basis elements, so $f = g$ by Lemma 27.4; uniqueness of the representation follows from the same lemma. ∎

Thus, once a basis is chosen, each alternating k-tensor can be written uniquely as $f = \sum_{[J]} d_J \psi_J$; the d_J are the **components** of f relative to $\{\psi_J\}$. Since each k-element subset of $\{1, \dots, n\}$ gives exactly one ascending k-tuple,

$$\dim \mathcal{A}^k(V) = \binom{n}{k} = \frac{n!}{k!(n-k)!}.$$

Theorem 27.6

Let $T : V \to W$ be a linear transformation. If f is an alternating tensor on W, then $T^* f$ is an alternating tensor on V.

Proof [Supplement]. $(T^*f)(\dots, \mathbf{v}_{i+1}, \mathbf{v}_i, \dots) = f(\dots, T\mathbf{v}_{i+1}, T\mathbf{v}_i, \dots) = -f(\dots, T\mathbf{v}_i, T\mathbf{v}_{i+1}, \dots) = -(T^*f)(\dots, \mathbf{v}_i, \mathbf{v}_{i+1}, \dots)$. ∎

27.3 Determinants

We now (at long last) construct the determinant for matrices larger than 3 by 3 (cf. §2).

Definition 27.5 (determinant)

Let $\phi_1, \dots, \phi_n$ be the basis of $\mathcal{L}^1(\mathbb{R}^n)$ dual to the usual basis. $\mathcal{A}^n(\mathbb{R}^n)$ has dimension 1; its unique elementary alternating n-tensor is $\psi_{1,\dots,n}$. For an n by n matrix $X = [\mathbf{x}_1 \ \cdots \ \mathbf{x}_n]$ define

$$\det X = \psi_{1,\dots,n}(\mathbf{x}_1, \dots, \mathbf{x}_n).$$

This satisfies the axioms of §2. Let $g(X) = \psi_I(\mathbf{x}_1, \dots, \mathbf{x}_n)$ with $I = (1, \dots, n)$; g is multilinear and alternating in the *columns* of X. So $f(A) = g(A^{\text{tr}})$ is multilinear and alternating in the rows of A, and $f(I_n) = g(I_n) = \psi_I(\mathbf{e}_1, \dots, \mathbf{e}_n) = 1$; hence f satisfies the axioms. By Theorem 2.11, $f(A) = f(A^{\text{tr}})$, so $f(A) = g(A)$ and g satisfies the axioms too. Theorem 27.5 gives the formula

$$\det X = \sum_\sigma (\operatorname{sgn}\sigma)\phi_I(\mathbf{x}_{\sigma(1)}, \dots, \mathbf{x}_{\sigma(n)}) = \sum_\sigma (\operatorname{sgn}\sigma) x_{1,\sigma(1)} \cdot x_{2,\sigma(2)} \cdots x_{n,\sigma(n)},$$

sometimes used as the definition of the determinant.

Theorem 27.7

Let ψ_I be an elementary alternating tensor on $\mathbb{R}^n$ corresponding to the usual basis for $\mathbb{R}^n$, where $I = (i_1, \dots, i_k)$. Given vectors $\mathbf{x}_1, \dots, \mathbf{x}_k$ of $\mathbb{R}^n$, let X be the matrix $X = [\mathbf{x}_1 \ \cdots \ \mathbf{x}_k]$. Then

$$\psi_I(\mathbf{x}_1, \dots, \mathbf{x}_k) = \det X_I,$$

where X_I denotes the matrix whose successive rows are rows $i_1, \dots, i_k$ of X.

Proof. $\psi_I(\mathbf{x}_1, \dots, \mathbf{x}_k) = \sum_\sigma (\operatorname{sgn}\sigma)\phi_I(\mathbf{x}_{\sigma(1)}, \dots, \mathbf{x}_{\sigma(k)}) = \sum_\sigma (\operatorname{sgn}\sigma) x_{i_1,\sigma(1)} \cdot x_{i_2,\sigma(2)} \cdots x_{i_k,\sigma(k)}$, which is the formula for $\det X_I$. ■

Example 27.2

The elementary alternating 3-tensors on $\mathbb{R}^4$ are

$$\psi_{i,j,k}(\mathbf{x}, \mathbf{y}, \mathbf{z}) = \det \begin{bmatrix} x_i & y_i & z_i \\ x_j & y_j & z_j \\ x_k & y_k & z_k \end{bmatrix},$$

for $(i, j, k) = (1, 2, 3), (1, 2, 4), (1, 3, 4), (2, 3, 4)$; the general element of $\mathcal{A}^3(\mathbb{R}^4)$ is a linear combination of these four.

Notation

For finite-dimensional V the exterior power $\Lambda^k(V^*)$ is naturally isomorphic to the alternating k-tensors, denoted $\Lambda^k(V^*)$ or $\Lambda^k(V)$ by various authors; we write $\mathcal{A}^k(V)$.

§28. The Wedge Product

Motivation. The wedge product makes the alternating tensors an algebra with basis dx_I.

$f \otimes g$ is almost never alternating. The properties matter more than the definition:

Theorem 28.1

Let V be a vector space. There is an operation that assigns, to each $f \in \mathcal{A}^k(V)$ and each $g \in \mathcal{A}^l(V)$, an element $f \wedge g \in \mathcal{A}^{k+l}(V)$, such that the following properties hold:

(1) (Associativity) $f \wedge (g \wedge h) = (f \wedge g) \wedge h$.

(2) (Homogeneity) $(cf) \wedge g = c(f \wedge g) = f \wedge (cg)$.

(3) (Distributivity) If f and g have the same order, $(f + g) \wedge h = f \wedge h + g \wedge h$ and $h \wedge (f + g) = h \wedge f + h \wedge g$.

(4) (Anticommutativity) If f and g have orders k and l, respectively, then $g \wedge f = (-1)^{kl} f \wedge g$.

(5) Given a basis $\mathbf{a}_1, \ldots, \mathbf{a}_n$ for V, let ϕ_i denote the dual basis for V^*, and let ψ_I denote the corresponding elementary alternating tensors. If $I = (i_1, \ldots, i_k)$ is an ascending k-tuple of integers from the set $\{1, \ldots, n\}$, then $\psi_I = \phi_{i_1} \wedge \phi_{i_2} \wedge \cdots \wedge \phi_{i_k}$.

These five properties characterize the product $\wedge$ uniquely for finite-dimensional spaces V. Furthermore, it has the following additional property:

(6) If $T : V \to W$ is a linear transformation, and if f and g are alternating tensors on W, then $T^*(f \wedge g) = T^* f \wedge T^* g$.

$f \wedge g$ is the **wedge product** of f and g. By (4), $f \wedge f = 0$ for an alternating tensor f of odd order.

Proof.

Step 1. For a k-tensor F on V (not necessarily alternating) define $A : \mathcal{L}^k(V) \to \mathcal{L}^k(V)$ by

$$AF = \sum_\sigma (\operatorname{sgn} \sigma) F^\sigma,$$

summed over $\sigma \in S_k$. (Some authors include a factor $1/k!$.) Thus $\psi_I = A\phi_I$. A has the properties:

(i) A is linear;

(ii) AF is alternating;

(iii) if F is already alternating, $AF = (k!)F$. Indeed (i) holds since $F \mapsto F^\sigma$ is linear;

(ii) by the computation $(AF)^\tau = \sum_\sigma (\operatorname{sgn} \sigma) F^{\tau \circ \sigma} = (\operatorname{sgn} \tau) \sum_\sigma (\operatorname{sgn} \tau \circ \sigma) F^{\tau \circ \sigma} = (\operatorname{sgn} \tau) AF$;

(iii) since $F^\sigma = (\operatorname{sgn} \sigma) F$ gives $AF = \sum_\sigma (\operatorname{sgn} \sigma)^2 F = (k!)F$.

Step 2. For $f \in \mathcal{A}^k(V)$, $g \in \mathcal{A}^l(V)$ define

$$f \wedge g = \frac{1}{k!\, l!} A(f \otimes g),$$

an alternating tensor of order $k+l$. *Motivation for* $1/k!l!$: in $(f \wedge g)(\mathbf{v}_1, \ldots, \mathbf{v}_{k+l}) = \frac{1}{k!l!} \sum_\sigma (\operatorname{sgn} \sigma) f(\mathbf{v}_{\sigma(1)}, \ldots, \mathbf{v}_{\sigma(k)})$ $g(\mathbf{v}_{\sigma(k+1)}, \ldots, \mathbf{v}_{\sigma(k+l)})$, permuting $\mathbf{v}_{\sigma(1)}, \ldots, \mathbf{v}_{\sigma(k)}$ among themselves and $\mathbf{v}_{\sigma(k+1)}, \ldots, \mathbf{v}_{\sigma(k+l)}$ among themselves gives $k!l!$ terms with the same value (since f, g are alternating, the sign changes cancel); dividing by $k!l!$ removes this redundancy.

Step 3. Homogeneity: $(cf) \wedge g = A((cf) \otimes g)/k!l! = A(c(f \otimes g))/k!l! = cA(f \otimes g)/k!l! = c(f \wedge g)$; similarly for the other part. Distributivity follows from distributivity of $\otimes$ and linearity of A.

Step 4. More generally, for tensors F, G of orders k, l (not necessarily alternating), $A(F \otimes G) = (-1)^{kl} A(G \otimes F)$. Let $\pi \in S_{k+l}$ with $(\pi(1), \ldots, \pi(k+l)) = (k+1, k+2, \ldots, k+l, 1, 2, \ldots, k)$; counting inversions, $\operatorname{sgn} \pi = (-1)^{kl}$. Since $(G \otimes F)^\pi(\mathbf{v}_1, \ldots, \mathbf{v}_{k+l}) = G(\mathbf{v}_{k+1}, \ldots, \mathbf{v}_{k+l}) \cdot F(\mathbf{v}_1, \ldots, \mathbf{v}_k)$, we have $(G \otimes F)^\pi = F \otimes G$. Hence

$$A(F \otimes G) = \sum_\sigma (\operatorname{sgn} \sigma)((G \otimes F)^\pi)^\sigma = (\operatorname{sgn} \pi) \sum_\sigma (\operatorname{sgn} \sigma \circ \pi)(G \otimes F)^{\sigma \circ \pi} = (\operatorname{sgn} \pi) A(G \otimes F),$$

since $\sigma \circ \pi$ runs over S_{k+l}. Taking $F = f$, $G = g$ gives anticommutativity.

Step 5. If F, G have orders k, l and $AF = 0$, then $A(F \otimes G) = 0$. In $A(F \otimes G)$, group together the terms with the same last factor $G(\mathbf{v}_{\sigma(k+1)}, \dots, \mathbf{v}_{\sigma(k+l)})$. They are

$$(\operatorname{sgn}\sigma)\Big[\sum_{\tau}(\operatorname{sgn}\tau)F(\mathbf{v}_{\sigma(\tau(1))}, \dots, \mathbf{v}_{\sigma(\tau(k))})\Big] \cdot G(\mathbf{v}_{\sigma(k+1)}, \dots, \mathbf{v}_{\sigma(k+l)}),$$

τ ranging over S_k, and the bracket is $AF(\mathbf{v}_{\sigma(1)}, \dots, \mathbf{v}_{\sigma(k)}) = 0$. Each group cancels.

Step 6. If F is any tensor and h alternating of order m, then $(AF) \wedge h = \frac{1}{m!}A(F \otimes h)$. With F of order k this reads $\frac{1}{k!m!}A((AF) \otimes h) = \frac{1}{m!}A(F \otimes h)$, i.e. (by linearity and distributivity) $A\{[AF - (k!)F] \otimes h\} = 0$. By Step 5 it suffices that $A[AF - (k!)F] = 0$, which follows from property (iii) of A, since AF is alternating: $A(AF) = (k!)AF$.

Step 7. For alternating f, g, h of orders k, l, m: $(f \wedge g) \wedge h = \frac{1}{k!l!m!}A((f \otimes g) \otimes h)$. Indeed with $F = f \otimes g$, $f \wedge g = \frac{1}{k!l!}AF$, so by Step 6, $(f \wedge g) \wedge h = \frac{1}{k!l!}(AF) \wedge h = \frac{1}{k!l!m!}A(F \otimes h)$.

Step 8. Associativity:

$$\begin{aligned}
(k!l!m!)(f \wedge g) \wedge h &= A((f \otimes g) \otimes h) && \text{by Step 7,}\\
&= A(f \otimes (g \otimes h)) && \text{by associativity of } \otimes,\\
&= (-1)^{k(l+m)}A((g \otimes h) \otimes f) && \text{by Step 4,}\\
&= (-1)^{k(l+m)}(l!m!k!)(g \wedge h) \wedge f && \text{by Step 7,}\\
&= (k!l!m!)f \wedge (g \wedge h) && \text{by anticommutativity.}
\end{aligned}$$

Step 9. Property (5). More generally, for 1-tensors $f_1, \dots, f_k$,

$$A(f_1 \otimes \cdots \otimes f_k) = f_1 \wedge \cdots \wedge f_k. \tag{$*$}$$

Then (5) follows since $\psi_I = A\phi_I = A(\phi_{i_1} \otimes \cdots \otimes \phi_{i_k})$. $(*)$ is trivial for $k = 1$; if true for $k-1$, let $F = f_1 \otimes \cdots \otimes f_{k-1}$; by Step 6 with $m = 1$, $A(F \otimes f_k) = (AF) \wedge f_k = (f_1 \wedge \cdots \wedge f_{k-1}) \wedge f_k$.

Step 10. Uniqueness: with only (1)–(5) one can compute wedge products in finite dimensions. Write $f = \sum_{[I]} b_I\psi_I$, $g = \sum_{[J]} c_J\psi_J$; distributivity and homogeneity give $f \wedge g = \sum_{[I]}\sum_{[J]} b_Ic_J\psi_I \wedge \psi_J$, and $\psi_I \wedge \psi_J = (\phi_{i_1} \wedge \cdots \wedge \phi_{i_k}) \wedge (\phi_{j_1} \wedge \cdots \wedge \phi_{j_l})$. By associativity and the rules $\phi_i \wedge \phi_j = -\phi_j \wedge \phi_i$, $\phi_i \wedge \phi_i = 0$ (from anticommutativity), $\psi_I \wedge \psi_J = 0$ if two indices coincide, and otherwise equals $\operatorname{sgn}\pi$ times ψ_K, where K is the ascending rearrangement of (I, J) and π the permutation effecting it.

Step 11. Property (6). For any tensor F on W, $T^*(F^\sigma) = (T^*F)^\sigma$; since T^* is linear, $T^*(AF) = A(T^*F)$. For alternating f, g of orders k, l on W,

$$T^*(f \wedge g) = \frac{1}{k!l!}T^*(A(f \otimes g)) = \frac{1}{k!l!}A(T^*(f \otimes g)) = \frac{1}{k!l!}A((T^*f) \otimes (T^*g)) = (T^*f) \wedge (T^*g),$$

using Theorem 26.5. ∎

Remark

With coefficient $1/(k+l)!$ instead of $1/k!l!$ the wedge product is still associative and Theorem 28.1 holds except (5), which gains a factor $k!$.

§29. Tangent Vectors and Differential Forms

As calculus has scalar and vector fields with ∇f, $\nabla \times \vec{F}$, $\nabla \cdot \vec{G}$, we introduce tensor fields, especially alternating ones (*differential forms*), and in §30 the operator d.

29.1 Tangent vectors and vector fields

Definition 29.1 (tangent vector, tangent space)

Given $\mathbf{x} \in \mathbb{R}^n$, a **tangent vector** to $\mathbb{R}^n$ at $\mathbf{x}$ is a pair $(\mathbf{x}; \mathbf{v})$ with $\mathbf{v} \in \mathbb{R}^n$. The tangent vectors at $\mathbf{x}$ form a vector space, the **tangent space** $\mathcal{T}_{\mathbf{x}}(\mathbb{R}^n)$, under

$$(\mathbf{x}; \mathbf{v}) + (\mathbf{x}; \mathbf{w}) = (\mathbf{x}; \mathbf{v} + \mathbf{w}), \qquad c(\mathbf{x}; \mathbf{v}) = (\mathbf{x}; c\mathbf{v}).$$

We picture $\mathbf{x}$ as a point and $\mathbf{v}$ as an arrow with initial point $\mathbf{x}$; $\mathcal{T}_{\mathbf{x}}(\mathbb{R}^n) = \mathbf{x} \times \mathbb{R}^n$. We do not add $(\mathbf{x}; \mathbf{v}) + (\mathbf{y}; \mathbf{w})$ if $\mathbf{x} \neq \mathbf{y}$.

Definition 29.2 (velocity vector)

Let (a, b) be an open interval and $\gamma : (a, b) \to \mathbb{R}^n$ of class C^r. The **velocity vector** of γ at parameter value t is $(\gamma(t); D\gamma(t))$.

For $\gamma(t) = x(t)\mathbf{e}_1 + y(t)\mathbf{e}_2 + z(t)\mathbf{e}_3$ in $\mathbb{R}^3$, calculus defines the velocity as $D\gamma(t) = \frac{dx}{dt}\mathbf{e}_1 + \frac{dy}{dt}\mathbf{e}_2 + \frac{dz}{dt}\mathbf{e}_3$.

Definition 29.3 (induced transformation α_*)

Let A be open in $\mathbb{R}^k$ or $\mathbb{H}^k$, $\alpha : A \to \mathbb{R}^n$ of class C^r, $\mathbf{x} \in A$, $\mathbf{p} = \alpha(\mathbf{x})$. The linear transformation $\alpha_* : \mathcal{T}_{\mathbf{x}}(\mathbb{R}^k) \to \mathcal{T}_{\mathbf{p}}(\mathbb{R}^n)$ **induced** by α is

$$\alpha_*(\mathbf{x}; \mathbf{v}) = (\mathbf{p}; D\alpha(\mathbf{x}) \cdot \mathbf{v}).$$

By the chain rule, $\alpha_*(\mathbf{x}; \mathbf{v})$ is the velocity vector of the curve $\gamma(t) = \alpha(\mathbf{x} + t\mathbf{v})$ at $t = 0$.

Lemma 29.1

Let A be open in $\mathbb{R}^k$ or $\mathbb{H}^k$; let $\alpha : A \to \mathbb{R}^m$ be of class C^r. Let B be an open set of $\mathbb{R}^m$ or $\mathbb{H}^m$ containing $\alpha(A)$; let $\beta : B \to \mathbb{R}^n$ be of class C^r. Then $(\beta \circ \alpha)_* = \beta_* \circ \alpha_*$.

Proof. With $\mathbf{y} = \alpha(\mathbf{x})$, $\mathbf{z} = \beta(\mathbf{y})$, the chain rule gives $(\beta \circ \alpha)_*(\mathbf{x}; \mathbf{v}) = (\mathbf{z}; D\beta(\mathbf{y}) \cdot D\alpha(\mathbf{x}) \cdot \mathbf{v}) = \beta_*(\mathbf{y}; D\alpha(\mathbf{x}) \cdot \mathbf{v}) = \beta_*(\alpha_*(\mathbf{x}; \mathbf{v}))$. ■

Definition 29.4 (tangent vector field)

If A is open in $\mathbb{R}^n$, a **tangent vector field** in A is a continuous $F : A \to \mathbb{R}^n \times \mathbb{R}^n$ with $F(\mathbf{x}) \in \mathcal{T}_{\mathbf{x}}(\mathbb{R}^n)$ for each $\mathbf{x}$; so $F(\mathbf{x}) = (\mathbf{x}; f(\mathbf{x}))$ for some $f : A \to \mathbb{R}^n$. If f is of class C^r, F is of class C^r.

Definition 29.5 (tangent space to a manifold)

Let M be a k-manifold of class C^r in $\mathbb{R}^n$, $\mathbf{p} \in M$, and $\alpha : U \to V$ a coordinate patch about $\mathbf{p}$ with $\alpha(\mathbf{x}) = \mathbf{p}$. The **tangent space to M at $\mathbf{p}$** is $\mathcal{T}_{\mathbf{p}}(M) = \alpha_*(\mathcal{T}_{\mathbf{x}}(\mathbb{R}^k))$, the set of vectors $\alpha_*(\mathbf{x}; \mathbf{v})$, $\mathbf{v} \in \mathbb{R}^k$.

$\mathcal{T}_{\mathbf{p}}(M)$ is a linear subspace of $\mathcal{T}_{\mathbf{p}}(\mathbb{R}^n)$, independent of the choice of α. It is spanned by the vectors $(\mathbf{p}; D\alpha(\mathbf{x}) \cdot \mathbf{e}_j) = (\mathbf{p}; \partial\alpha/\partial x_j)$, $j = 1, \ldots, k$, which are independent since $D\alpha$ has rank k; so they form a basis. The union of the $\mathcal{T}_{\mathbf{p}}(M)$ is the **tangent bundle** $\mathcal{T}(M)$; a **tangent vector field to** M is a continuous $F : M \to \mathcal{T}(M)$ with $F(\mathbf{p}) \in \mathcal{T}_{\mathbf{p}}(M)$.

29.2 Tensor fields

Definition 29.6 (tensor field, differential form)

Let A be open in $\mathbb{R}^n$. A **k-tensor field** in A is a function ω assigning to each $\mathbf{x} \in A$ a k-tensor $\omega(\mathbf{x}) \in \mathcal{L}^k(\mathcal{T}_{\mathbf{x}}(\mathbb{R}^n))$, such that $\omega(\mathbf{x})((\mathbf{x};\mathbf{v}_1),\dots,(\mathbf{x};\mathbf{v}_k))$ is continuous as a function of $(\mathbf{x},\mathbf{v}_1,\dots,\mathbf{v}_k)$; if it is of class C^r, ω is a tensor field of class C^r. If $\omega(\mathbf{x})$ is alternating for each $\mathbf{x}$, ω is a **differential form** (or simply a **form**) of order k on A. More generally, a k-tensor field on an m-manifold M assigns to each $\mathbf{p} \in M$ an element of $\mathcal{L}^k(\mathcal{T}_{\mathbf{p}}(M))$; if alternating, it is a differential form on M.

A tensor field on an open set of $\mathbb{R}^n$ containing M restricts to one on M; conversely any tensor field on M extends (non-trivially). We use tensor fields on open sets of $\mathbb{R}^n$.

Definition 29.7 (elementary forms)

$(\mathbf{x};\mathbf{e}_1),\dots,(\mathbf{x};\mathbf{e}_n)$ is the **usual basis** of $\mathcal{T}_{\mathbf{x}}(\mathbb{R}^n)$. The **elementary 1-forms** $\tilde{\phi}_i$ on $\mathbb{R}^n$ are defined by $\tilde{\phi}_i(\mathbf{x})(\mathbf{x};\mathbf{e}_j) = 0$ if $i \neq j$ and 1 if $i = j$. For an ascending k-tuple $I = (i_1,\dots,i_k)$, the **elementary k-form** is $\tilde{\psi}_I(\mathbf{x}) = \tilde{\phi}_{i_1}(\mathbf{x}) \wedge \cdots \wedge \tilde{\phi}_{i_k}(\mathbf{x})$.

For each $\mathbf{x}$, $\tilde{\phi}_1(\mathbf{x}),\dots,\tilde{\phi}_n(\mathbf{x})$ is the basis of $\mathcal{L}^1(\mathcal{T}_{\mathbf{x}}(\mathbb{R}^n))$ dual to the usual basis, and $\tilde{\psi}_I(\mathbf{x})$ is the corresponding elementary alternating tensor. They are of class C^∞, since

$$\tilde{\phi}_i(\mathbf{x})(\mathbf{x};\mathbf{v}) = v_i, \qquad \tilde{\psi}_I(\mathbf{x})((\mathbf{x};\mathbf{v}_1),\dots,(\mathbf{x};\mathbf{v}_k)) = \det X_I, \quad X = [\mathbf{v}_1 \ \cdots \ \mathbf{v}_k].$$

A k-form on A can be written uniquely as $\omega(\mathbf{x}) = \sum_{[I]} b_I(\mathbf{x})\tilde{\psi}_I(\mathbf{x})$; the scalar functions b_I are the **components** of ω relative to the standard elementary forms.

Lemma 29.2

Let ω be a k-form on the open set A of $\mathbb{R}^n$. Then ω is of class C^r if and only if its component functions b_I are of class C^r on A.

Proof. Write $\omega = \sum_{[I]} b_I\tilde{\psi}_I$. The $\tilde{\psi}_I$ are C^∞, so if the b_I are C^r, so is ω. Conversely, if ω is C^r as a function of $(\mathbf{x},\mathbf{v}_1,\dots,\mathbf{v}_k)$, then for ascending J the function $\omega(\mathbf{x})((\mathbf{x};\mathbf{e}_{j_1}),\dots,(\mathbf{x};\mathbf{e}_{j_k}))$ is C^r in $\mathbf{x}$; it equals $b_J(\mathbf{x})$. ■

Lemma 29.3

Let ω and η be k-forms, and let θ be an l-form, on the open set A of $\mathbb{R}^n$. If ω and η and θ are of class C^r, so are $a\omega + b\eta$ and $\omega \wedge \theta$.

Proof. $a\omega + b\eta$ is a linear combination of C^r functions. Write $\omega = \sum_{[I]} b_I\tilde{\psi}_I$, $\theta = \sum_{[J]} c_J\tilde{\psi}_J$; then $\omega \wedge \theta = \sum_{[I]}\sum_{[J]} b_Ic_J\tilde{\psi}_I \wedge \tilde{\psi}_J$. To express this in elementary forms, drop terms with repeated indices, rewrite the rest with indices in ascending order, and collect terms: each component of $\omega \wedge \theta$ is a sum, with signs ± 1, of functions b_Ic_J, hence of class C^r (Lemma 29.2). ■

29.3 Differential forms of order zero

Definition 29.8 (scalar field; 0-form)

If A is open in $\mathbb{R}^n$ and $f : A \to \mathbb{R}$ is of class C^r, f is a **scalar field** in A, also called a **differential form of order** 0. The wedge product of 0-forms is $f \wedge g = f \cdot g$, and of a 0-form with a k-form is $(\omega \wedge f)(\mathbf{x}) = (f \wedge \omega)(\mathbf{x}) = f(\mathbf{x}) \cdot \omega(\mathbf{x})$.

All formal algebraic properties of the wedge product hold; anticommutativity holds since 0-forms have order 0: $f \wedge g = (-1)^0 g \wedge f$, $f \wedge \omega = (-1)^0 \omega \wedge f$.

Convention

Henceforth, Roman letters such as f, g, h denote 0-forms, and Greek letters such as ω, η, θ denote k-forms for $k > 0$.

§30. The Differential Operator

Motivation. The exterior derivative d unifies grad, curl and div, and $d^2 = 0$ contains $\operatorname{curl}\operatorname{grad} = 0$, $\operatorname{div}\operatorname{curl} = 0$.

The operator d carries k-forms to $(k+1)$-forms. We first define it on 0-forms.

30.1 The differential of a 0-form

Definition 30.1 (differential of a function)

Let A be open in $\mathbb{R}^n$ and $f : A \to \mathbb{R}$ of class C^r. The 1-form df on A is

$$df(\mathbf{x})(\mathbf{x}; \mathbf{v}) = f'(\mathbf{x}; \mathbf{v}) = Df(\mathbf{x}) \cdot \mathbf{v}.$$

It is the **differential** of f; it is of class C^{r-1} as a function of $\mathbf{x}$ and $\mathbf{v}$.

Theorem 30.1

The operator d is linear on 0-forms.

Proof. If $h = af + bg$, then $Dh(\mathbf{x}) = aDf(\mathbf{x}) + bDg(\mathbf{x})$, so $dh(\mathbf{x})(\mathbf{x}; \mathbf{v}) = a\, df(\mathbf{x})(\mathbf{x}; \mathbf{v}) + b\, dg(\mathbf{x})(\mathbf{x}; \mathbf{v})$. ■

Lemma 30.2

Let $\tilde{\phi}_1, \ldots, \tilde{\phi}_n$ be the elementary 1-forms in $\mathbb{R}^n$. Let $\pi_i : \mathbb{R}^n \to \mathbb{R}$ be the ith projection function, defined by the equation $\pi_i(x_1, \ldots, x_n) = x_i$. Then $d\pi_i = \tilde{\phi}_i$.

Proof. $d\pi_i$ is a C^∞ 1-form, and $d\pi_i(\mathbf{x})(\mathbf{x}; \mathbf{v}) = D\pi_i(\mathbf{x}) \cdot \mathbf{v} = [0 \cdots 0\ 1\ 0 \cdots 0] \cdot \mathbf{v} = v_i$. ■

Convention

If $\mathbf{x}$ denotes the general point of $\mathbb{R}^n$, we denote the ith projection function $\mathbb{R}^n \to \mathbb{R}$ by x_i. Then dx_i equals the elementary 1-form $\tilde{\phi}_i$ in $\mathbb{R}^n$. If $I = (i_1, \ldots, i_k)$ is an ascending k-tuple, we write $dx_I = dx_{i_1} \wedge \cdots \wedge dx_{i_k}$ for the elementary k-form $\tilde{\psi}_I$. The general k-form is uniquely

$\omega = \sum_{[I]} b_I\, dx_I$. For an arbitrary k-tuple J we also set $dx_J = dx_{j_1} \wedge \cdots \wedge dx_{j_k}$; note dx_J is not the differential of a form but a wedge product of elementary 1-forms.

These forms are characterized by

$$dx_i(\mathbf{x})(\mathbf{x};\mathbf{v}) = v_i, \qquad dx_I(\mathbf{x})((\mathbf{x};\mathbf{v}_1),\dots,(\mathbf{x};\mathbf{v}_k)) = \det X_I, \quad X = [\mathbf{v}_1 \cdots \mathbf{v}_k].$$

(Using x_i for the function π_i confuses a function with its value—an abuse as common as speaking of "the function $x^3 + 2x + 1$".)

Theorem 30.3

Let A be open in $\mathbb{R}^n$; let $f : A \to \mathbb{R}$ be of class C^r. Then

$$df = (D_1 f)\, dx_1 + \cdots + (D_n f)\, dx_n.$$

In particular, $df = 0$ if f is a constant function.

In Leibniz notation, $df = \frac{\partial f}{\partial x_1} dx_1 + \cdots + \frac{\partial f}{\partial x_n} dx_n$—a formula in calculus books whose meaning is not explained there.

Proof. Evaluate both sides on $(\mathbf{x};\mathbf{v})$: $df(\mathbf{x})(\mathbf{x};\mathbf{v}) = Df(\mathbf{x}) \cdot \mathbf{v}$, while $\sum_i D_i f(\mathbf{x})\, dx_i(\mathbf{x})(\mathbf{x};\mathbf{v}) = \sum_i D_i f(\mathbf{x}) v_i$. ■

Convention

Since df is only C^{r-1} when f is C^r, to avoid counting degrees of differentiability we henceforth restrict ourselves to manifolds, maps, vector fields, and forms that are of class C^∞.

30.2 The differential of a k-form

Definition 30.2 ($\Omega^k(A)$)

For A open in $\mathbb{R}^n$, $\Omega^k(A)$ is the vector space of all (C^∞) k-forms on A, the **linear space of k-forms** on A.

Theorem 30.4

Let A be an open set in $\mathbb{R}^n$. There exists a unique linear transformation

$$d : \Omega^k(A) \to \Omega^{k+1}(A),$$

defined for $k \geq 0$, such that:

(1) If f is a 0-form, then df is the 1-form $df(\mathbf{x})(\mathbf{x};\mathbf{v}) = Df(\mathbf{x}) \cdot \mathbf{v}$.

(2) If ω and η are forms of orders k and l, respectively, then $d(\omega \wedge \eta) = d\omega \wedge \eta + (-1)^k \omega \wedge d\eta$.

(3) For every form ω, $d(d\omega) = 0$.

d is the **differential operator**, and $d\omega$ the **differential** of ω.

Proof.

Step 1. Uniqueness. First, (2) and (3) imply $d(d\omega_1 \wedge \cdots \wedge d\omega_k) = 0$ for any forms $\omega_1, \dots, \omega_k$: for $k = 1$ this is (3); assuming it for $k-1$ and setting $\eta = d\omega_2 \wedge \cdots \wedge d\omega_k$, (2) gives $d(d\omega_1 \wedge \eta) =$

$d(d\omega_1)\wedge\eta \pm d\omega_1\wedge d\eta$, both terms vanishing. Since d is linear, it suffices to compute d on $\omega = f\,dx_I$:

$$d\omega = d(f\wedge dx_I) = df\wedge dx_I + f\wedge d(dx_I) = df\wedge dx_I,$$

since $dx_I = dx_{i_1}\wedge\cdots\wedge dx_{i_k}$ with $dx_{i_j} = d(x_{i_j})$. So $d\omega$ is determined by d on 0-forms, which is (1).

Step 2. Definition. For $\omega = \sum_{[I]} f_I\,dx_I$ define $d\omega = \sum_{[I]} df_I\wedge dx_I$. It is C^∞: $d\omega = \sum_{[I]}[\sum_{j=1}^n (D_j f_I)dx_j]\wedge dx_I$; deleting terms with j among the indices of I, rearranging the rest into ascending order and collecting terms, each component of $d\omega$ is a linear combination of the $D_j f_I$. (If ω were only C^r, $d\omega$ would be C^{r-1}.) d is linear: for $\omega = \sum f_I dx_I$, $\eta = \sum g_I dx_I$, $d(a\omega+b\eta) = \sum d(af_I + bg_I)\wedge dx_I = \sum(a\,df_I + b\,dg_I)\wedge dx_I = a\,d\omega + b\,d\eta$.

Step 3. For an *arbitrary* k-tuple J, $d(f\,dx_J) = df\wedge dx_J$. If two indices of J coincide, $dx_J = 0$. Otherwise let I be the ascending rearrangement of J by the permutation π; by anticommutativity $dx_I = (\operatorname{sgn}\pi)dx_J$, and since d is linear and $\wedge$ homogeneous, $d(f\,dx_I) = df\wedge dx_I$ gives $(\operatorname{sgn}\pi)d(f\,dx_J) = (\operatorname{sgn}\pi)df\wedge dx_J$.

Step 4. Property (2) for $k = l = 0$: $d(f\wedge g) = \sum_j D_j(f\cdot g)dx_j = \sum_j (D_j f)g\,dx_j + \sum_j f(D_j g)dx_j = df\wedge g + f\wedge dg$.

Step 5. Property (2) in general. For $k, l > 0$, by linearity take $\omega = f\,dx_I$, $\eta = g\,dx_J$:

$$\begin{aligned} d(\omega\wedge\eta) &= d(fg\,dx_I\wedge dx_J) = d(fg)\wedge dx_I\wedge dx_J && \text{by Step 3,}\\ &= (df\,g + f\,dg)\wedge dx_I\wedge dx_J && \text{by Step 4,}\\ &= (df\wedge dx_I)\wedge(g\,dx_J) + (-1)^k(f\,dx_I)\wedge(dg\wedge dx_J) = d\omega\wedge\eta + (-1)^k\omega\wedge d\eta, \end{aligned}$$

the sign arising because dx_I is a k-form and dg a 1-form. If k or l is 0 the same argument works with dx_I or dx_J absent.

Step 6. If f is a 0-form, $d(df) = 0$:

$$d(df) = d\Big(\sum_j D_j f\,dx_j\Big) = \sum_j d(D_j f)\wedge dx_j = \sum_j\sum_i D_i D_j f\,dx_i\wedge dx_j = \sum_{i<j}(D_i D_j f - D_j D_i f)\,dx_i\wedge dx_j,$$

which vanishes by equality of mixed partials (Theorem 6.3).

Step 7. If ω is a k-form, $k > 0$, $d(d\omega) = 0$. By linearity take $\omega = f\,dx_I$: $d(d\omega) = d(df\wedge dx_I) = d(df)\wedge dx_I - df\wedge d(dx_I)$ by (2). $d(df) = 0$ by Step 6, and $d(dx_I) = d(1)\wedge dx_I = 0$ by definition. ■

Definition 30.3 (closed and exact forms)

Let A be open in $\mathbb{R}^n$. A 0-form f on A is **exact** on A if it is constant on A; a k-form ω with $k > 0$ is **exact** on A if $\omega = d\theta$ for some $(k-1)$-form θ on A. A k-form ω ($k \geq 0$) is **closed** on A if $d\omega = 0$.

Every exact form is closed: $df = 0$ for constant f, and $d(d\theta) = 0$. Conversely every closed form on A is exact if $A = \mathbb{R}^n$, or more generally if A is "star-convex" (Chapter 8), but not in general. If every closed k-form on A is exact, A is **homologically trivial in dimension** k.

Example 30.1

Let $A = \{(x,y)\in\mathbb{R}^2 : x \neq 0\}$ and $f(x,y) = x/|x|$. f is C^∞ on A and $df = 0$, but f is not exact on A since it is not constant.

Example 30.2

In differential equations, $P(x,y)\,dx + Q(x,y)\,dy = 0$ is called exact if there is f with $P = \partial f/\partial x$, $Q = \partial f/\partial y$; in our terms the 1-form $P\,dx + Q\,dy$ is the differential of f, hence exact. In $\mathbb{R}^3$, a vector field $\vec{F} = P\vec{i} + Q\vec{j} + R\vec{k}$ is *conservative* if it is the gradient of a scalar field f; this is the same as saying the form $P\,dx + Q\,dy + R\,dz$ is the differential of the 0-form f.

§31. Application to Vector and Scalar Fields

Linear maps between $\Omega^k(A)$ and the C^∞ vector fields on A show that forms generalize vector analysis in $\mathbb{R}^3$ (used in §38).

Definition 31.1 (gradient, divergence)

Let A be open in $\mathbb{R}^n$, $f : A \to \mathbb{R}$ a scalar field. The **gradient** of f is the vector field

$$(\operatorname{grad} f)(\mathbf{x}) = (\mathbf{x}; D_1 f(\mathbf{x})\mathbf{e}_1 + \cdots + D_n f(\mathbf{x})\mathbf{e}_n).$$

If $G(\mathbf{x}) = (\mathbf{x}; g(\mathbf{x}))$ is a vector field with $g(\mathbf{x}) = g_1(\mathbf{x})\mathbf{e}_1 + \cdots + g_n(\mathbf{x})\mathbf{e}_n$, the **divergence** of G is the scalar field

$$(\operatorname{div} G)(\mathbf{x}) = D_1 g_1(\mathbf{x}) + \cdots + D_n g_n(\mathbf{x}).$$

Theorem 31.1

Let A be an open set in $\mathbb{R}^n$. There exist vector space isomorphisms α_i and β_j as in the following diagram:

$$\begin{array}{ccc} \text{Scalar fields in } A & \xrightarrow{\alpha_0} & \Omega^0(A) \\ {\scriptstyle \operatorname{grad}}\downarrow & & \downarrow{\scriptstyle d} \\ \text{Vector fields in } A & \xrightarrow{\alpha_1} & \Omega^1(A) \end{array} \qquad \begin{array}{ccc} \text{Vector fields in } A & \xrightarrow{\beta_{n-1}} & \Omega^{n-1}(A) \\ {\scriptstyle \operatorname{div}}\downarrow & & \downarrow{\scriptstyle d} \\ \text{Scalar fields in } A & \xrightarrow{\beta_n} & \Omega^n(A) \end{array}$$

such that $d \circ \alpha_0 = \alpha_1 \circ \operatorname{grad}$ and $d \circ \beta_{n-1} = \beta_n \circ \operatorname{div}$.

Proof. For scalar fields f, h and vector fields $F(\mathbf{x}) = (\mathbf{x}; \sum f_i(\mathbf{x})\mathbf{e}_i)$, $G(\mathbf{x}) = (\mathbf{x}; \sum g_i(\mathbf{x})\mathbf{e}_i)$ define

$$\alpha_0 f = f, \quad \alpha_1 F = \sum_{i=1}^n f_i\,dx_i, \quad \beta_{n-1} G = \sum_{i=1}^n (-1)^{i-1} g_i\,dx_1 \wedge \cdots \wedge \widehat{dx_i} \wedge \cdots \wedge dx_n, \quad \beta_n h = h\,dx_1 \wedge \cdots \wedge dx_n,$$

where $\widehat{a}$ means a is omitted. [Supplement]Each is linear, and bijective because $\{dx_i\}$, $\{dx_1 \wedge \cdots \wedge \widehat{dx_i} \wedge \cdots \wedge dx_n\}$ and $\{dx_1 \wedge \cdots \wedge dx_n\}$ are the elementary bases of Ω^1, Ω^{n-1}, Ω^n (Lemma 29.2). By Theorem 30.3, $d(\alpha_0 f) = df = \sum_i D_i f\,dx_i = \alpha_1(\operatorname{grad} f)$. And

$$d(\beta_{n-1} G) = \sum_i (-1)^{i-1} \sum_j D_j g_i\,dx_j \wedge dx_1 \wedge \cdots \wedge \widehat{dx_i} \wedge \cdots \wedge dx_n = \sum_i (-1)^{i-1} D_i g_i (-1)^{i-1} dx_1 \wedge \cdots \wedge dx_n = \beta_n(\operatorname{div} G),$$

since only $j = i$ survives and moving dx_i past $i-1$ factors gives the sign $(-1)^{i-1}$. ∎

For $\mathbb{R}^3$ there is also a *curl* operator.

Definition 31.2 (curl)

Let A be open in $\mathbb{R}^3$ and $F(\mathbf{x}) = (\mathbf{x}; \sum f_i(\mathbf{x})\mathbf{e}_i)$ a vector field. The **curl** of F is

$$(\operatorname{curl} F)(\mathbf{x}) = (\mathbf{x}; (D_2 f_3 - D_3 f_2)\mathbf{e}_1 + (D_3 f_1 - D_1 f_3)\mathbf{e}_2 + (D_1 f_2 - D_2 f_1)\mathbf{e}_3),$$

remembered as the symbolic determinant $\det \begin{bmatrix} \mathbf{e}_1 & \mathbf{e}_2 & \mathbf{e}_3 \\ D_1 & D_2 & D_3 \\ f_1 & f_2 & f_3 \end{bmatrix}$.

Theorem 31.2

Let A be an open set in $\mathbb{R}^3$. There exist vector space isomorphisms α_i and β_j as in the following diagram:

$$\begin{array}{ccc} \text{Scalar fields in } A & \xrightarrow{\alpha_0} & \Omega^0(A) \\ \downarrow{\scriptstyle \text{grad}} & & \downarrow{\scriptstyle d} \\ \text{Vector fields in } A & \xrightarrow{\alpha_1} & \Omega^1(A) \\ \downarrow{\scriptstyle \text{curl}} & & \downarrow{\scriptstyle d} \\ \text{Vector fields in } A & \xrightarrow{\beta_2} & \Omega^2(A) \\ \downarrow{\scriptstyle \text{div}} & & \downarrow{\scriptstyle d} \\ \text{Scalar fields in } A & \xrightarrow{\beta_3} & \Omega^3(A) \end{array}$$

such that $d \circ \alpha_0 = \alpha_1 \circ \text{grad}$, $d \circ \alpha_1 = \beta_2 \circ \text{curl}$, and $d \circ \beta_2 = \beta_3 \circ \text{div}$.

Proof. The maps are those of Theorem 31.1 with $n = 3$: $\beta_2 G = g_1\, dx_2 \wedge dx_3 - g_2\, dx_1 \wedge dx_3 + g_3\, dx_1 \wedge dx_2$. Only the middle equation needs checking [Supplement]:

$$\begin{aligned} d(\alpha_1 F) &= d(f_1 dx_1 + f_2 dx_2 + f_3 dx_3) = \sum_{i,j} D_j f_i\, dx_j \wedge dx_i \\ &= (D_2 f_3 - D_3 f_2)\, dx_2 \wedge dx_3 + (D_3 f_1 - D_1 f_3)\, dx_3 \wedge dx_1 + (D_1 f_2 - D_2 f_1)\, dx_1 \wedge dx_2, \end{aligned}$$

and since $dx_3 \wedge dx_1 = -dx_1 \wedge dx_3$, this is $\beta_2(\operatorname{curl} F)$. ∎

Consequently, $d(d\omega) = 0$ translates into $\operatorname{curl}\operatorname{grad} f = 0$ and $\operatorname{div}\operatorname{curl} F = 0$ in $\mathbb{R}^3$.

§32. The Action of a Differentiable Map

A C^∞ map $\alpha : A \to \mathbb{R}^n$, A open in $\mathbb{R}^k$, induces α_* on tangent spaces, and linear maps T induce dual maps T^* on alternating tensors. Combined, α induces α^* on forms, preserving linear structure, $\wedge$ and d.

Definition 32.1 (pullback α^*)

Let A be open in $\mathbb{R}^k$, $\alpha : A \to \mathbb{R}^m$ of class C^∞, and B open in $\mathbb{R}^m$ containing $\alpha(A)$. The **dual transformation of forms** $\alpha^* : \Omega^l(B) \to \Omega^l(A)$ is defined as follows: for a 0-form f on B, $(\alpha^* f)(\mathbf{x}) = f(\alpha(\mathbf{x}))$; for an l-form ω on B, $l > 0$,

$$(\alpha^*\omega)(\mathbf{x})((\mathbf{x}; \mathbf{v}_1), \ldots, (\mathbf{x}; \mathbf{v}_l)) = \omega(\alpha(\mathbf{x}))(\alpha_*(\mathbf{x}; \mathbf{v}_1), \ldots, \alpha_*(\mathbf{x}; \mathbf{v}_l)).$$

Since $f, \omega, \alpha, D\alpha$ are C^∞, so are $\alpha^* f$ and $\alpha^*\omega$. If α is constant, $\alpha^* f$ is constant and $\alpha^*\omega$ is the zero tensor. If $\alpha(\mathbf{x}) = \mathbf{y}$ and $T = \alpha_* : \mathcal{T}_{\mathbf{x}}(\mathbb{R}^k) \to \mathcal{T}_{\mathbf{y}}(\mathbb{R}^m)$, then $T^*(\omega(\mathbf{y})) = (\alpha^*\omega)(\mathbf{x})$, since

$$T^*(\omega(\mathbf{y}))((\mathbf{x}; \mathbf{v}_1), \ldots, (\mathbf{x}; \mathbf{v}_l)) = \omega(\alpha(\mathbf{x}))(\alpha_*(\mathbf{x}; \mathbf{v}_1), \ldots, \alpha_*(\mathbf{x}; \mathbf{v}_l)).$$

So earlier results on dual transformations become results about forms:

Theorem 32.1

Let A be open in $\mathbb{R}^k$; let $\alpha : A \to \mathbb{R}^m$ be a C^∞ map. Let B be open in $\mathbb{R}^m$ and contain $\alpha(A)$; let $\beta : B \to \mathbb{R}^n$ be a C^∞ map. Let ω, η, θ be forms defined in an open set C of $\mathbb{R}^n$ containing $\beta(B)$; assume ω and η have the same order. The transformations α^* and β^* have the following properties:
(1) $\beta^*(a\omega + b\eta) = a(\beta^*\omega) + b(\beta^*\eta)$.
(2) $\beta^*(\omega \wedge \theta) = \beta^*\omega \wedge \beta^*\theta$.
(3) $(\beta \circ \alpha)^*\omega = \alpha^*(\beta^*\omega)$.

Proof. For forms of positive order, (1) and (3) restate Theorem 26.5 (using Lemma 29.1 for (3)), and (2) restates property (6) of Theorem 28.1. [Supplement](Order zero.) If ω, η are 0-forms, $\beta^*(af+bg) = (af+bg)\circ\beta = a(f\circ\beta)+b(g\circ\beta)$. If $\theta = g$ is a 0-form, $\beta^*(\omega\wedge g)(\mathbf{y}) = g(\beta(\mathbf{y}))\cdot\beta^*\omega(\mathbf{y}) = (\beta^*\omega \wedge \beta^*g)(\mathbf{y})$, and similarly if ω is a 0-form. For 0-forms, (3) reads $f \circ (\beta \circ \alpha) = (f \circ \beta) \circ \alpha$. ■

We now compute $\alpha^*\omega$ when ω is a 1-form and when ω is a k-form on $\mathbb{R}^n$ with $\alpha : A \to \mathbb{R}^n$, A open in $\mathbb{R}^k$; since α^* is linear, preserves wedge products and $\alpha^* f = f \circ \alpha$, it suffices to treat elementary forms.

Theorem 32.2

Let A be open in $\mathbb{R}^k$; let $\alpha : A \to \mathbb{R}^n$ be a C^∞ map. Let $\mathbf{x}$ denote the general point of $\mathbb{R}^k$; let $\mathbf{y}$ denote the general point of $\mathbb{R}^n$. Then dx_i and dy_i denote the elementary 1-forms in $\mathbb{R}^k$ and $\mathbb{R}^n$, respectively.
(a) $\alpha^*(dy_i) = d\alpha_i$.
(b) If $I = (i_1, \ldots, i_k)$ is an ascending k-tuple from the set $\{1, \ldots, n\}$, then

$$\alpha^*(dy_I) = \left(\det \frac{\partial \alpha_I}{\partial \mathbf{x}}\right) dx_1 \wedge \cdots \wedge dx_k, \qquad \text{where} \quad \frac{\partial \alpha_I}{\partial \mathbf{x}} = \frac{\partial(\alpha_{i_1}, \ldots, \alpha_{i_k})}{\partial(x_1, \ldots, x_k)}.$$

Proof. (a) With $\mathbf{y} = \alpha(\mathbf{x})$,

$$(\alpha^*(dy_i))(\mathbf{x})(\mathbf{x};\mathbf{v}) = dy_i(\mathbf{y})(\alpha_*(\mathbf{x};\mathbf{v})) = i\text{th component of } D\alpha(\mathbf{x})\cdot\mathbf{v} = \sum_{j=1}^{k} D_j\alpha_i(\mathbf{x})v_j = \sum_{j=1}^{k} \frac{\partial \alpha_i}{\partial x_j}(\mathbf{x})\, dx_j(\mathbf{x})(\mathbf{x};\mathbf{v}).$$

So $\alpha^*(dy_i) = \sum_j \frac{\partial \alpha_i}{\partial x_j} dx_j$, which equals $d\alpha_i$ by Theorem 30.3.

(b) $\alpha^*(dy_I)$ is a k-form on an open set of $\mathbb{R}^k$, so $\alpha^*(dy_I) = h\, dx_1 \wedge \cdots \wedge dx_k$ for a scalar function h, found by evaluating on $((\mathbf{x};\mathbf{e}_1), \ldots, (\mathbf{x};\mathbf{e}_k))$:

$$h(\mathbf{x}) = dy_I(\mathbf{y})(\alpha_*(\mathbf{x};\mathbf{e}_1), \ldots, \alpha_*(\mathbf{x};\mathbf{e}_k)) = dy_I(\mathbf{y})((\mathbf{y};\partial\alpha/\partial x_1), \ldots, (\mathbf{y};\partial\alpha/\partial x_k)) = \det[D\alpha(\mathbf{x})]_I = \det \frac{\partial \alpha_I}{\partial \mathbf{x}}.$$

■

By (a), $\alpha^*(dy_i)$ is dy_i with $y_i = \alpha_i(\mathbf{x})$ substituted; computing $\alpha^*(dy_I) = d\alpha_{i_1} \wedge \cdots \wedge d\alpha_{i_k}$ directly is laborious for $k > 2$.

Theorem 32.3

Let A be open in $\mathbb{R}^k$; let $\alpha : A \to \mathbb{R}^n$ be of class C^∞. If ω is an l-form defined in an open set of $\mathbb{R}^n$ containing $\alpha(A)$, then

$$\alpha^*(d\omega) = d(\alpha^*\omega).$$

Proof. Let $\mathbf{x}$, $\mathbf{y}$ be the general points of $\mathbb{R}^k$, $\mathbb{R}^n$.

Step 1. For a 0-form f:

$$\alpha^*(df) = \alpha^*\Big(\sum_{i=1}^{n}(D_i f)\,dy_i\Big) = \sum_{i=1}^{n}((D_i f)\circ\alpha)\,d\alpha_i. \tag{$*$}$$

On the other hand

$$d(\alpha^* f) = d(f\circ\alpha) = \sum_{j=1}^{k} D_j(f\circ\alpha)\,dx_j. \tag{$**$}$$

By the chain rule, $D(f\circ\alpha)(\mathbf{x}) = Df(\mathbf{y})\cdot D\alpha(\mathbf{x})$, so $D_j(f\circ\alpha) = \sum_i((D_i f)\circ\alpha)\cdot D_j\alpha_i$. Substituting in $(**)$,

$$d(\alpha^* f) = \sum_{j=1}^{k}\sum_{i=1}^{n}((D_i f)\circ\alpha)D_j\alpha_i\,dx_j = \sum_{i=1}^{n}((D_i f)\circ\alpha)\,d\alpha_i, \tag{$***$}$$

which agrees with $(*)$.

Step 2. For forms of positive order, by linearity it suffices to take $\omega = f\,dy_I$ with I ascending. Then

$$\alpha^*(d\omega) = \alpha^*(df\wedge dy_I) = \alpha^*(df)\wedge\alpha^*(dy_I), \tag{$\dagger$}$$

$$d(\alpha^*\omega) = d[(\alpha^* f)\wedge\alpha^*(dy_I)] = d(\alpha^* f)\wedge\alpha^*(dy_I) + (\alpha^* f)\wedge 0, \tag{$\dagger\dagger$}$$

since $d(\alpha^*(dy_I)) = d(d\alpha_{i_1}\wedge\cdots\wedge d\alpha_{i_l}) = 0$. Compare $(\dagger)$ and $(\dagger\dagger)$ using Step 1. ■

With this, the algebra of differential forms and d, as summarized in §30 and this section, is all we need.

Forms versus vector fields

A vector field on A cannot be pushed to $\alpha(A)$: if $\alpha(\mathbf{x}_1) = \alpha(\mathbf{x}_2) = \mathbf{y}$, $\mathbf{x}_1\neq\mathbf{x}_2$, F gives two vectors $\alpha_*F(\mathbf{x}_1)$, $\alpha_*F(\mathbf{x}_2)$ at $\mathbf{y}$. For a diffeomorphism, $G(\mathbf{y}) = \alpha_*(F(\alpha^{-1}(\mathbf{y})))$ works but need not commute with grad, curl, div or the operators α_i, β_j of §31. Forms pull back under *any* C^∞ map, compatibly with $\wedge$ and d.